

Topics for the video

21 August 2026 13:13

First, we will understand the background of ORPO.

Second, we will discuss what ORPO is and why it was introduced.

Third, we will compare ORPO with techniques like DPO, PPO, and GRPO.

Fourth, we will understand the mathematics behind ORPO in a simple way.

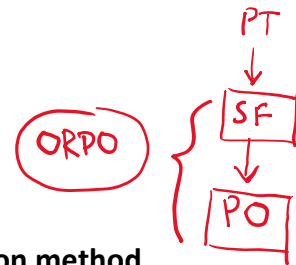
Fifth, we will discuss the limitations of ORPO.

finally, we will move to the practical implementation

- ORPO: Odd Ratio Preference optimization
- Research paper of ORPO: <https://arxiv.org/pdf/2403.07691>
- ORPO was introduced by Jiwoo Hong, Noah Lee, and James Thorne in 2024
- It was introduced in Korea Advanced Institute of Science and Technology(KAIST AI) it is a research university in **South Korea**
- Jiwoo Hong currently: **AI Engineer at LinkedIn**
- James Thorne: faculty member at KAIST's
- The ORPO paper first appeared on arXiv in **March 2024** and was later published at **EMNLP 2024**.
- **EMNLP** stands for the **Conference on Empirical Methods in Natural Language Processing**. It is a major international research conference in the fields of NLP and Large Language Models.
- EMNLP 2024 was held in **Miami, USA, from November 12 to 16, 2024**.
- The Main Conference acceptance rate approximately **20%**, roughly one out of every five submitted papers accepting.

Monolithic Preference Optimization technique without Reference Model

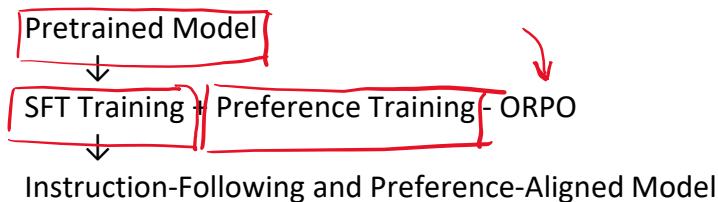
ORPO is a single-stage preference optimization method



monolithic preference optimization or Single Stage preference optimization method

meaning is instruction learning and preference learning are integrated into one objective and in one training process.

ORPO combines Supervised Fine-Tuning & Preference Alignment into a **single training stage**.



ORPO directly trains Pretrained Model using chosen and rejected responses.

preference data for ORPO:

Prompt | Chosen Response | Rejected Response

It teaches the same model to:

1. Learn the chosen response through normal SFT loss.
2. Prefer the chosen response over the rejected response through an Odds Ratio loss.

ORPO

ORPO Training Flow

Prompt + Chosen Response + Rejected Response

↓
Single Policy Model

↓
Calculate Chosen Response Probability

↓
Calculate Rejected Response Probability

↓
SFT Loss on the Chosen Response

+
Odds Ratio Loss between Chosen and Rejected

↓
Combined ORPO Loss

↓
Update the Same Model

ORPO does not require:

- Separate Reward Model
- Value/Critic Model
- PPO or reinforcement learning
- Frozen Reference Model

ORPO: Research Results and Real-World Usage

The original ORPO experiments trained several models, including:

- OPT models from 125M to 1.3B
- Phi-2 2.7B
- Llama-2 7B
- Mistral 7B
- Researchers publicly released the Mistral-ORPO-Alpha and Mistral-ORPO-Beta checkpoints.
- Publicly documented ORPO usage is currently seen mainly in:
 - Open-source models
 - Research models
 - Domain-specific experiments
 - custom or internal fine-tuning use cases
- ORPO has also been used in actual projects and publicly released models, so it is not limited to the research experiments.
- A strong example is Hugging Face H4's Zephyr-ORPO-141B-A35B. It was built from the Mixtral-8x22B base model and aligned with ORPO to create a helpful assistant. According to its model card, it was trained on 7,000 preference examples using 32 H100 GPUs for approximately 1.3 hours.
- ORPO has also been applied to large models and domain-specific applications:
 - ORPO has been used in biomedical research, Pharma research, 32B web-navigation model(web navigation / web-agent tasks).

But For general commercial preference fine-tuning, DPO has stronger publicly available production evidence. For example, OpenAI's Preference Fine-Tuning API officially uses DPO.

Why Were Preference Optimization Algorithms Like DPO and ORPO Introduced, and What Limitations of Standard SFT Do They Address?

In normal SFT, the model is trained only on the correct or chosen response.

Example:

Prompt:

How should I treat a mild fever?

Target Response:

Rest, stay hydrated, monitor your temperature,
and contact a doctor if symptoms become severe.

The SFT objective increases the probability of the target response.

However, SFT does not explicitly tell the model:

This response is preferred.

This other response is rejected.

The ORPO researchers observed that while SFT increases the probability of chosen responses, it may also increase the probability of rejected responses because both responses can contain similar language, tokens, topics and writing patterns.

SFT has no direct mechanism for penalizing the rejected answer.

ORPO solves this by adding a relatively small preference penalty to the standard SFT loss.

Difference between RLHF(PPO) vs GRPO vs DPO vs ORPO

Method	Reward Model/Function	Critic	Reference Model	Multiple Generated Responses	SFT Integrated
PPO-RLHF	yes	Yes	Yes	Yes - but not multiple	No
GRPO	Reward function/model	No separate Critic	Yes	Yes	No
DPO	No separate RM	No	Yes	No online generation required	No
ORPO	No separate RM	No	No	No online generation required	Yes

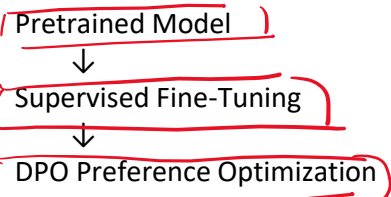
DPO: Remove Reward Model, Critic and PPO loop.

DPO Simplified Preference Learning

ORPO: Also remove the frozen Reference Model and merge SFT with preference optimization.

However, DPO generally still uses:

1. A model already trained through SFT (Policy Model)
2. A trainable A frozen Reference Model
3. A separate preference-optimization stage



ORPO asks: Can we perform supervised fine-tuning and preference alignment together, without using a Reference Model?
The answer is **yes**.

DPO → Policy Model + Reference Model

ORPO → Policy Model only

What Does "Reference-Model-Free" Mean?

In DPO, two models are used:

Policy Model → Trainable
Reference Model → Frozen

The Reference Model acts as a baseline and limits how far the Policy Model moves from its original behavior.

A Reward Model does not directly provide a score; instead, the reward signal is automatically derived by comparing the probabilities of the Policy Model and the Reference Model.

Chosen implicit reward high
Rejected implicit reward low



Loss low

Rejected implicit reward high
Chosen implicit reward low



Loss high



Model is corrected

ORPO uses only one trainable Policy Model. It does not require an additional frozen Reference Model during preference training.

Therefore, ORPO requires less memory and computation than DPO.

ORPO does not use a Reward Model or a Reference Model. The preference signal is derived by comparing the **odds of chosen and rejected responses from the same Policy Model**, while SFT loss trains the chosen response.

Chosen odds high —
Rejected odds low —



OR loss low

Rejected odds high
Chosen odds low



OR loss high



Model is corrected

In the current industry and open-source ecosystem, DPO is the most widely used and safest for preference tuning. For reasoning models, GRPO/GSPO/RLVR are more relevant and rapidly growing techniques. ORPO is conceptually simple, but its adoption is significantly behind DPO.

Rank	Technique	Real-world position
1	<u>DPO</u>	<u>Most common offline pairwise preference tuning</u>
2	<u>GRPO/RLVR/GSPO</u>	<u>Very popular for reasoning and verifiable tasks</u>
3	PPO-based RLHF	<u>Still important in advanced and hybrid frontier pipelines</u>
4	<u>KTO/SimPO and variants</u>	<u>Used for specific data or training requirements</u>
5	<u>ORPO</u>	<u>Simple and reference-free, but comparatively behind</u>

Companies such as Meta, Alibaba, Microsoft, Mistral AI and Hugging Face have publicly used DPO in the post-training of models such as Llama 3, Qwen2.5-VL, Phi-3, Mixtral Instruct and Zephyr.

Organization	Model	How DPO was used
<u>Meta</u>	<u>Llama 3.1 / Llama 3.2 Instruct</u>	<u>Post-training used multiple rounds of SFT + Rejection Sampling + DPO. (Meta AI)</u>
Alibaba Cloud – Qwen Team	Qwen2.5-VL	Used offline DPO for preference alignment, followed by online GRPO. (Qwen)
Microsoft	Phi-3-mini / Phi-3.5-Vision	Post-training included SFT followed by DPO for chat, reasoning, safety and multimodal alignment. (arXiv)
Mistral AI	Mixtral 8x7B Instruct	Optimized through SFT and <u>DPO</u> for better instruction following. (Mistral AI)
Hugging Face H4	<u>Zephyr-7B-beta</u>	Mistral-7B was fine-tuned using preference data with DPO. (Hugging Face)

Question: What is 2 + 2?

Two answers were given:

Chosen answer: 4**Rejected answer:** 5

ORPO teaches the model two things:

Learn to write the correct answer "4"

Understand that "4" is better than "5"

That is why: $L_{ORPO} = L_{SFT} + \lambda L_{OR}$

In ORPO, the SFT loss teaches the model to generate the chosen answer, while the Odds Ratio loss teaches it to rank the chosen answer higher than the rejected answer.

$$L_{ORPO} = L_{SFT} + \lambda L_{OR}$$

- L_{SFT} : Teaches the model to generate the chosen response.
- L_{OR} : Teaches the model to prefer the chosen response over the rejected response.
- λ : Controls how much importance is given to the preference loss.

 λ controls how much importance preference learning receives.

- Chosen score higher $\rightarrow$ OR loss small
- Rejected score higher $\rightarrow$ OR loss large

$$L_{ORPO} = L_{SFT} + \lambda \left[-\log \sigma \left(\log \frac{\text{odds}(y_w | x)}{\text{odds}(y_l | x)} \right) \right]$$

Probability is converted into odds:

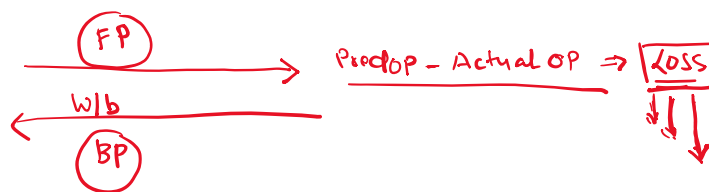
$$\text{odds}(P) = \frac{P}{1 - P}$$

$$S_{\text{chosen}} = \log(\text{chosen odds})$$

$$S_{\text{rejected}} = \log(\text{rejected odds})$$

$$\Delta_{OR} = S_{\text{chosen}} - S_{\text{rejected}}$$

$$L_{OR} = -\log \sigma(\Delta_{OR})$$

Loss is small $\rightarrow$ the model is performing wellLoss is large $\rightarrow$ the model is performing incorrectly

Part 1: What does SFT Loss do?

~~SFT tells the model: "Learn to generate the chosen answer."~~

Suppose the model assigns these scores:

$$P(\text{chosen}) = 0.75$$

$$P(\text{rejected}) = 0.25$$

This means the model assigns a ~~75% score~~ to the ~~chosen answer~~ and a ~~25% score~~ to the ~~rejected answer~~.

SFT loss:

$$L_{\text{SFT}} = -\log P(\text{chosen})$$

$$L_{\text{SFT}} = -\log(0.75) \approx 0.288$$

The probability of the chosen answer is good.

Remember only this:

Chosen probability high $\rightarrow$ SFT loss low

Chosen probability low $\rightarrow$ ~~SFT loss high~~

SFT loss teaches the model to write the chosen answer.

Part 2: What are Odds?

This is the most confusing part. But It is very simple.

~~Probability: $P = 0.75$~~

This means:

Chance of the answer occurring = 75

Chance of the answer not occurring = 25

Odds ask:

"How much greater is the chance of occurring than the chance of not occurring?"

Formula:

$$\text{odds}(P) = \frac{P}{1 - P}$$

Chosen answer odds:

$$\text{odds}_{\text{chosen}} = \frac{0.75}{1 - 0.75}$$

$$= \frac{0.75}{0.25} = 3$$

This means: The odds of the chosen answer are 3:1.

Rejected answer odds:

$$\begin{aligned}\text{odds}_{\text{rejected}} &= \frac{0.25}{1 - 0.25} \\ &= \frac{0.25}{0.75} = \frac{1}{3} \quad \text{1:3}\end{aligned}$$

This means the rejected answer has a much lower score.

Part 3: Comparing the Chosen and Rejected Answers

Now ORPO asks: How much greater are the odds of the chosen answer than the odds of the rejected answer?

$$\begin{aligned}\text{Odds Ratio} &= \frac{\text{Chosen Odds}}{\text{Rejected Odds}} \\ &= \frac{3}{1/3} = 9\end{aligned}$$

This means: The odds of the chosen response are 9 times the odds of the rejected response.

Remember: this does not mean the probability is 9 times greater. It is the ratio of the odds.

Part 4: Why do we apply Log?

Instead of directly using the large ratio, ORPO takes its log:

$$\log(9) \approx 2.197$$

The simple purpose of log:

Convert large ratios into manageable numbers.

Log also converts division into subtraction:

$$\begin{aligned}\log\left(\frac{\text{Chosen Odds}}{\text{Rejected Odds}}\right) \\ = \log(\text{Chosen Odds}) - \log(\text{Rejected Odds})\end{aligned}$$

Therefore, we write:

$$S_{\text{chosen}} = \log(\text{Chosen Odds})$$

$$S_{\text{rejected}} = \log(\text{Rejected Odds})$$

Then:

$$\Delta_{\text{OR}} = S_{\text{chosen}} - S_{\text{rejected}}$$

In our example:

$$\Delta_{\text{OR}} = 2.197$$

Meaning of Delta:

$$\Delta_{OR} = 2.197$$

Meaning of Delta:

Delta positive → chosen is better

Delta zero → both are equal

Delta negative → rejected is better

Part 5: What does Sigmoid do?

Now we have:

$$\Delta_{OR} = 2.197$$

Sigmoid converts any number into a value between 0 and 1:

$$\sigma(\Delta) = \frac{1}{1 + e^{-\Delta}}$$

$$\sigma(2.197) = 0.9$$

[0-1]

$$\left\{ \begin{array}{l} P(C) = 0.75 \\ P(R) = 0.25 \end{array} \right\}$$

Think of sigmoid as a judge meter:

Value close to 1 → chosen is strongly better

Value 0.5 → both are equal

Value close to 0 → rejected is better

In our example: 0.9

This means the ORPO objective strongly prefers the chosen answer.

It is not necessary to call this calibrated real-world confidence; it is better to understand it as a preference score.

Part 6: What does OR Loss do?

OR loss:

$$L_{OR} = -\log \sigma(\Delta_{OR})$$

In our example, the sigmoid value was 0.9:

$$L_{OR} = -\log(0.9)$$

$$L_{OR} \approx 0.105$$

The loss is small because the model is correctly ranking the chosen answer above the rejected answer.

Simple rule of negative log:

Sigmoid close to 1 → loss close to 0

Sigmoid close to 0 → loss becomes very large

This means:

Small punishment for correct ranking, large punishment for wrong ranking.

Part 7: Final ORPO Loss

Now we combine both losses:

$$L_{ORPO} = L_{SFT} + \lambda L_{OR}$$

We have:

$$L_{SFT} = 0.288$$

$$L_{OR} = 0.105$$

Suppose:

$$\lambda = 0.1$$

Then:

$$L_{ORPO} = 0.288 + 0.1(0.105)$$

$$= 0.288 + 0.0105$$

$$L_{ORPO} \approx 0.2985$$

What does Lambda do?

Think of λ as a volume button.

$$L_{ORPO} = L_{SFT} + \lambda L_{OR}$$

Lambda small $\rightarrow$ more focus on generating the chosen answer

Lambda large $\rightarrow$ more focus on comparing the chosen and rejected answers

Lambda decides how loud the preference learning signal will be.

What happens if the model ranks them differently?

Now suppose the values are reversed:

$$P(\text{chosen}) = 0.25$$

$$P(\text{rejected}) = 0.75$$

Now the rejected answer receives the higher score.

Chosen odds:

$$\frac{0.25}{0.75} = \frac{1}{3}$$

Rejected odds:

$$\frac{0.75}{0.25} = 3$$

Odds ratio:

$$\frac{1/3}{3} = \frac{1}{9}$$

Log:

$$\log(1/9) \approx -2.197$$

Sigmoid:

$$\sigma(-2.197) = 0.1$$

OR loss:

$$-\log(0.1) \approx 2.303$$

OR LOSS:

$$-\log(0.1) \approx 2.303$$

Now the loss is very large.

This means the model will receive a strong correction:

"You ranked the ~~rejected~~ answer as ~~better~~. Correct it."

Top 5 Major Benefits of ORPO

1. No Separate Reward Model
 2. No PPO or Reinforcement Learning Loop
 3. No Frozen Reference Model
 4. Single Training Stage
 5. Lower Memory Requirement Than DPO
- 

Top 5 Important Limitations of ORPO



1. Not Validated Beyond 7B in the Original Paper
2. Limited Comparison with Other Preference-Optimization Methods
3. Limited Domain and Dataset Generalization Evidence
4. Sensitive to Noisy or Low-Quality Preference Data
5. SimPO: Simple Preference Optimization with a Reference-Free Reward
Not Consistently Better Than Newer Methods such as SimPO

<https://arxiv.org/html/2405.14734v1>